

Effective Duration of a Zero-Coupon Bond



The Origin of Confusion

"Effective Duration of a zero-coupon bond equals its maturity"

Common Misreading

"% price change = T% for a 1% change in yield"

This is incorrect!

Core Misunderstanding

Mixing two different concepts:

Duration (sensitivity measure)

Actual % price change

What Duration Actually Measures

DURATION APPROXIMATION

$$\% \Delta P \approx -D \cdot \Delta y$$

1

Rate of Change

Duration is the slope, not the actual % change itself

2

Units: "Years"

But meaning = interest rate sensitivity per unit yield change

!

Interpretation

"% price change per 1 unit (100%) change in yield"

Effective Duration Formula

$$D_{\text{eff}} = \frac{P_- - P_+}{2P_0\Delta y}$$

P_- Price if yield decreases

P_+ Price if yield increases

Δy Small yield change

KEY INSIGHT

Link to Calculus

Effective duration approximates the derivative:

$$D \approx -\frac{1}{P} \frac{dP}{dy}$$

This is the first derivative of price with respect to yield

Zero-Coupon Bond Pricing and Taylor Expansion

ZCB PRICE FUNCTION

Price of a Zero-Coupon Bond

$$P(y) = \frac{1}{(1 + y)^T}$$

- Only one cash flow at maturity
- Entire price driven by discounting
- No intermediate coupons

MATHEMATICAL FOUNDATION

Taylor Series Approximation

$$f(x + h) \approx f(x) + f'(x) \cdot h$$

- First-order Taylor expansion
- Ignores higher-order terms
- Converts a curve into a straight line

Mathematical Derivation: Taylor to Duration

Apply Taylor to ZCB price function

$$1 \quad (1 + y \pm \Delta y)^{-T} \approx (1 + y)^{-T} \left(1 \mp \frac{T \Delta y}{1 + y} \right)$$

Substitute into effective duration formula

$$2 \quad P_- \approx P_0 \left(1 + \frac{T \Delta y}{1 + y} \right) \quad P_+ \approx P_0 \left(1 - \frac{T \Delta y}{1 + y} \right)$$

Simplify the numerator

$$3 \quad P_- - P_+ = 2P_0 \frac{T \Delta y}{1 + y}$$

FINAL DERIVATION

$$4 \quad D_{\text{eff}} = \frac{2P_0 \frac{T \Delta y}{1 + y}}{2P_0 \Delta y} = \frac{T}{1 + y}$$

KEY RESULT

Comparing Duration Formulas for ZCB

EFFECTIVE DURATION

$$D_{\text{eff}} = \frac{T}{1 + y}$$

Adjusts for current yield level. Always slightly less than T.

MACAULAY DURATION

$$D_{\text{mac}} = T$$

Weighted average time to receive cash flows. Exactly equals maturity for ZCB.

Numerical Example: 5-Year ZCB at 1% Yield

SETUP

Maturity	Yield	Shift
T = 5	y = 1%	100 bps

EFFECTIVE DURATION

$$D_{\text{eff}} = \frac{1 - 0.9057}{2 \times 0.9515 \times 0.01}$$

$$= 4.95$$

PRICE CALCULATIONS

$$P_0 = \frac{1}{(1.01)^5} = 0.9515$$

$$P_- = \frac{1}{(1.00)^5} = 1.0000$$

$$P_+ = \frac{1}{(1.02)^5} = 0.9057$$

DURATION ESTIMATE

$$-D \times \Delta y$$

$$= -4.95 \times 0.01$$

$$= -4.95\%$$

ACTUAL CHANGE

$$\frac{P_+ - P_0}{P_0}$$

$$= \frac{0.9057 - 0.9515}{0.9515}$$

$$= -4.81\%$$

Why the Gap? Convexity

Duration estimate

-4.95%

Actual change

-4.81%

WHY THE DIFFERENCE?

Price-yield relationship is convex
Duration assumes a linear relationship

Error arises because:

Taylor expansion ignores higher-order terms (convexity)

FULL TAYLOR EXPANSION

$$f(x + h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + \dots$$

1

First-order term

= Duration

2

Second-order term

= Convexity (ignored in duration)

Key Takeaways

1

Effective Duration Formula for a ZCB

$$D_{\text{eff}} = \frac{T}{1 + y}$$

2

"Duration = Maturity" Clarified

Exact for Macaulay, approximate for effective

3

Actual Price Change

Differs from estimate due to convexity

FINAL INTUITION

A zero-coupon bond has **all its value at maturity**, so its time exposure is T .

Effective duration **adjusts this for discounting** and approximates price sensitivity using a linear model.

This is why it is **close to** — but **not exactly** — T .